SOC542 Statistical Methods in Sociology II

Categorical outcomes

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Course updates

- Homework 4 due Friday, 4/17 at 5pm
- Project preliminary results due next Friday, 4/24 at 5pm
  - See last week's slides for details
- Presentations in class on 5/7

Course updates

- Labs
  - This week is final lab on lecture material
  - Remaining labs are project workshops
    - Attendance is still mandatory, but use the time to develop your projects and troubleshoot

Plan

- Categorical outcomes
- Multinomial logistic regression
- Ordered logistic regression

Categorical outcomes

Categories of categories

- A categorical outcome consists of *three or more discrete categories*
- *Ordered* categorical outcomes
  - e.g. Likert scales: Strongly agree, agree, neither agree nor disagree, disagree, strongly disagree
- *Unordered* (or nominal) categorical outcomes
  - e.g. Single, in a relationship, married, divorced, it's complicated.

Categorical outcomes

Intervals

- If a categorical variable is *ordered* then there should be an **interval** between categories such that each category can be positioned on a single dimension.
  - These intervals may vary between categories:
    - e.g. The difference between good and very good may be larger than difference between good and okay.
- Categories without any ordering do not have clearly defined intervals between categories.

Categorical outcomes

Modeling categories using existing approaches

- OLS regression
  - Only suitable if there are many categories and intervals are *evenly spaced*
- *One-versus-rest* logistic regression models
  - One model for each category with a binary outcome
  - Limitations: Loss of information

Data

GSS 2018

- Two outcomes from the GSS 2018:
  - Unordered: Marital status
    - Married, widowed, divorced, separated, never married
  - Ordered: Self-reported health
    - Excellent, Good, Fair, Poor

Models for categorical outcomes

- We will be considering two different approaches using variations of logistic regression:
  1. Unordered outcomes modeled using **multinomial** logistic regression
  2. Ordered outcomes modeled using **ordinal** logistic regression

Multinomial logistic regression

- **Multinomial logistic regression** generalizes logistic regression to *unordered* categorical outcomes.
- For a set of K outcomes, we can model the linear propensity for outcome k using a linear model with m predictors.

$$\lambda_k = \beta_{0k} + \beta_{1k}x_1 + ... + \beta_{mk}x_m$$

- Jointly estimate a set of equations, one for each category.

Multinomial logistic regression

- The probability of outcome y_k is represented by the **softmax** link function.[1] The probability of outcome k is the exponentiated linear propensity of outcome k relative to the sum of exponentiated linear propensities of all outcomes in the set K (Kruschke 2015: 650).

$$P(y = k|X) = \text{softmax}_K(\lambda_k) = \frac{e^{\lambda_k}}{\sum_{i \in K} e^{\lambda_i}}$$

[1] The approach is therefore sometimes referred to as **softmax regression**.

Multinomial logistic regression

- Because the probabilities across categories must sum to one, one category will always produce the following equation:

$$\lambda_r = \beta_{0r} + \beta_{1r}x_1 + ... + \beta_{mr}x_m = 0 + 0x_1 + ... + 0x_m = 0$$

- We therefore select a category to leave out as the *reference category*.
- Estimated coefficients can thus be interpreted as the relative log odds of each outcome compared to the reference category.

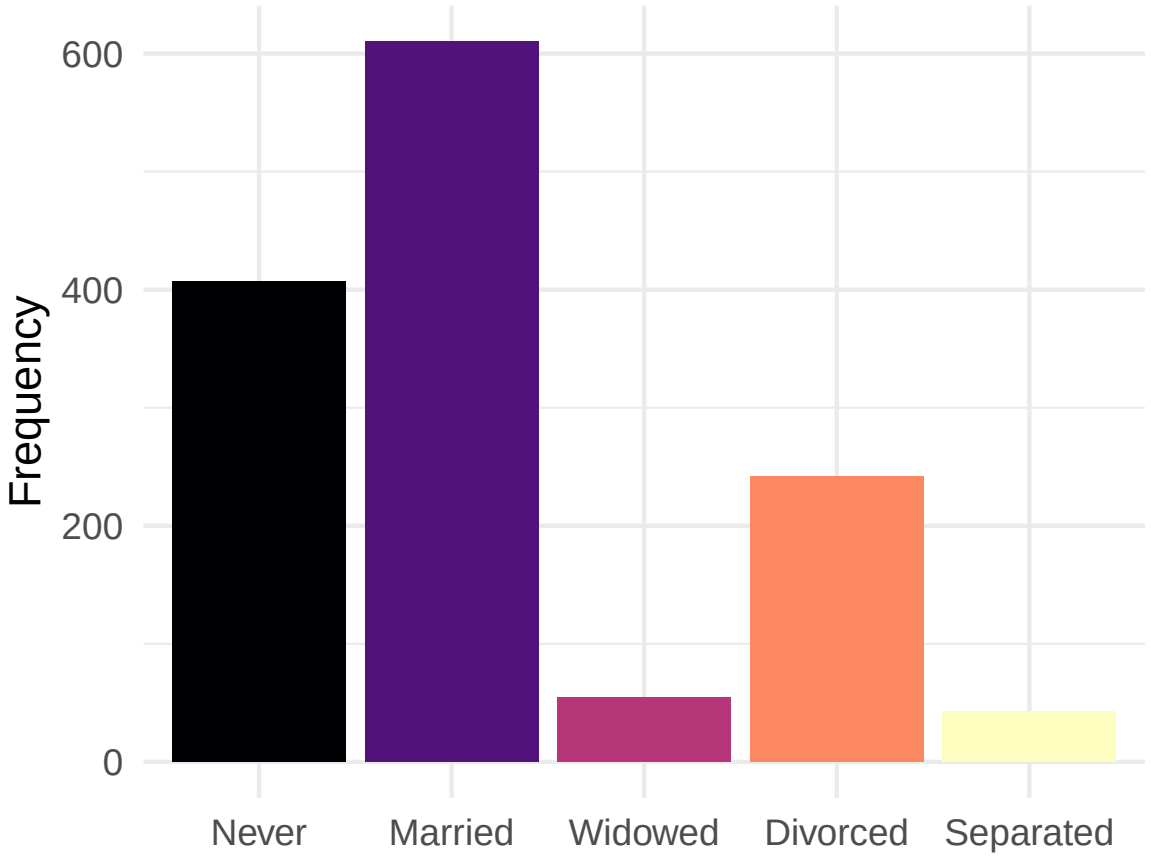
Multinomial logistic regression

Estimation

- These models are more complex than other GLMs due to the estimation of multiple equations.
- Maximum likelihood models can be estimated using `nnet::multinom`[2]
- Bayesian models can be estimated using the `brms` package and `family = categorical(link = "logit")`.

[2] Other packages are available but require additional data manipulation before modeling. See this blog for further discussion.

Data: Marital status



Multinomial logistic regression

Estimation

```
library(nnet)
gss$marital <- relevel(gss$marital, ref = "Never")
m1 <- multinom(marital ~ age + sex + log(realrinc) + educ, data = gss)
```

```
## # weights:  30 (20 variable)
## initial  value 2184.007247
## iter  10 value 1667.335362
## iter  20 value 1459.416635
## iter  30 value 1441.935116
## final  value 1441.935011
## converged
```

Multinomial logistic regression

	(1)			
	Married	Widowed	Divorced	Separated
(Intercept)	-6.546***	-10.986***	-8.047***	-7.817***
	(0.669)	(1.500)	(0.860)	(1.544)
age	0.092***	0.187***	0.122***	0.096***
	(0.007)	(0.015)	(0.008)	(0.014)
sexMale	-0.365*	-1.422***	-0.900***	-0.689*
	(0.153)	(0.347)	(0.196)	(0.347)
log(realrinc)	0.385***	0.262*	0.413***	0.500**
	(0.069)	(0.128)	(0.087)	(0.167)
educ	-0.014	-0.125*	-0.081*	-0.197***
	(0.028)	(0.058)	(0.035)	(0.055)
Num.Obs.	1357			
R2	0.162			

Multinomial logistic regression

	(1)			
	Married	Widowed	Divorced	Separated
(Intercept)	0.001***	0.000***	0.000***	0.000***
	(0.001)	(0.000)	(0.000)	(0.001)
age	1.096***	1.206***	1.130***	1.100***
	(0.007)	(0.018)	(0.009)	(0.015)
sexMale	0.694*	0.241***	0.407***	0.502*
	(0.107)	(0.084)	(0.080)	(0.174)
log(realrinc)	1.470***	1.299*	1.511***	1.649**
	(0.102)	(0.166)	(0.131)	(0.275)
educ	0.987	0.882*	0.922*	0.821***
	(0.027)	(0.051)	(0.032)	(0.045)
Num.Obs.	1357			
R2	0.162			

Multinomial logistic regression

Interpretation

- Each column is an equation for a specified outcome category compared to the *baseline* (Never married).
- For example, the first column represents the following equation:

$$log(\frac{P(y = \text{married})}{P(y = \text{never married})})$$

$$= \beta_{10} + \beta_{11} Age + \beta_{12} Sex + \beta_{13} Income + \beta_{14} Educ$$

Multinomial logistic regression

Interpretation

- β_{11} indicates that a one-year increase in age is associated with a .092 change in the log odds of being married relative to never married.
- Unlike standard logistic regression, where e^{β} is an odds ratio, e^{β} in a multinomial model is a **relative risk ratio**: $e^{\beta_{11}} = 1.096$ is the multiplicative change in $P(y = \text{married})/P(y = \text{never married})$ for a one-unit increase in age.
  - The probability of being married *relative to never married* increases by 9.6% each year

Multinomial logistic regression

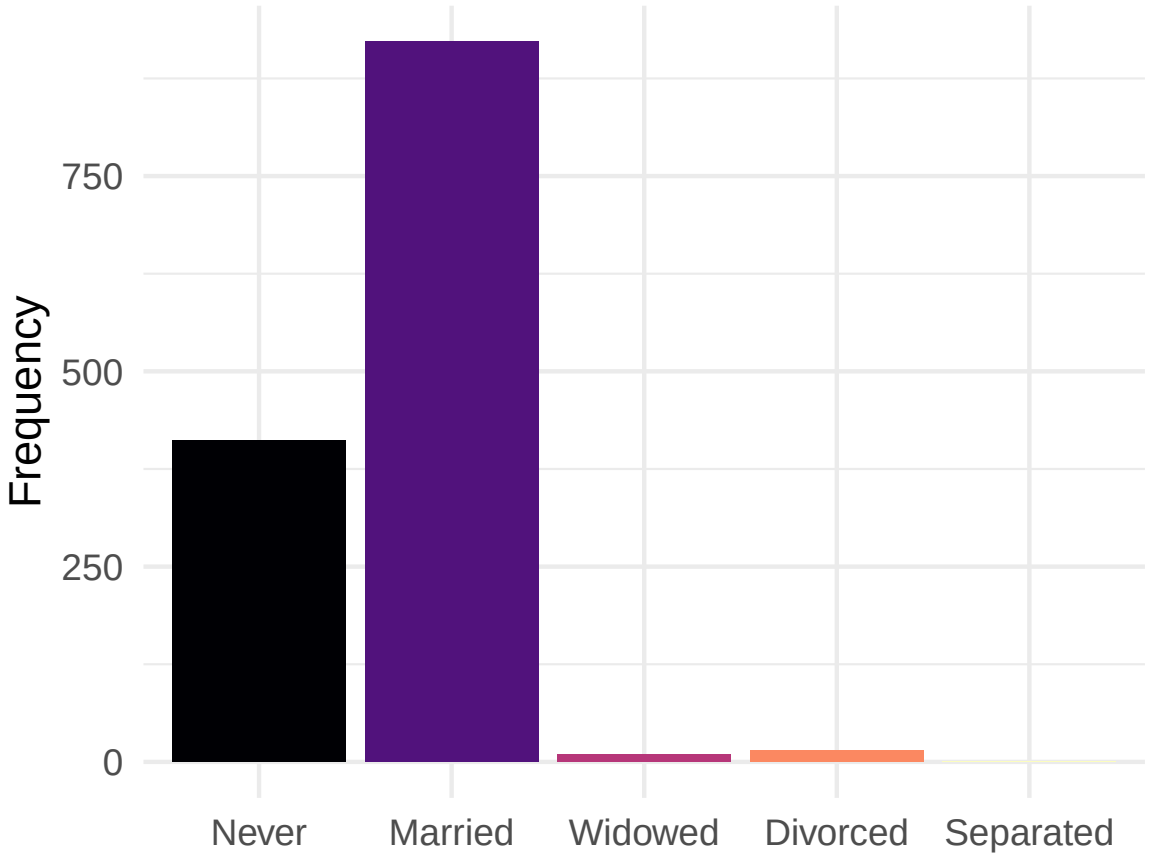
Predictions

The `predict` function returns a factor variable containing the highest probability category for each observation.

```
preds <- predict(m1, gss %>%
                     drop_na(age, sex, realrinc, educ, marital)
                 )
preds %>% head(20)
```

```
##  [1] Married  Married  Married  Married  Divorced Married  Married
##  [9] Married  Widowed  Married  Married  Married  Married  Married
## [17] Divorced Never    Married  Married
## Levels: Never Married Widowed Divorced Separated
```

Multinomial logistic regression



Multinomial logistic regression

Predictions

- The model predicts almost all people as never married or married.
- It rarely predicts widowed or divorced and did not predict any people to be separated.
- Data imbalances make never/married the most likely categories.
  - Additional variables would likely help to improve accuracy for other categories.

Multinomial logistic regression

Predictions

Setting `type = "probs"` returns a matrix of predicted probabilities for each observation. Each element indicates $P(y_i = k)$.

```
probs <- predict(m1, type = "probs", gss %>% drop_na())
probs %>% round(3) %>% head(5)
```

```
##   Never Married Widowed Divorced Separated
## 1 0.052   0.459   0.117    0.331     0.042
## 2 0.278   0.522   0.010    0.148     0.042
## 3 0.059   0.692   0.025    0.205     0.019
## 4 0.215   0.611   0.007    0.140     0.027
## 5 0.008   0.265   0.370    0.328     0.029
```

Multinomial logistic regression

Predictions

The probabilities for each observation sum to one, a feature of the softmax function.

```
probs %>% head(5) %>% rowSums() %>% as.numeric()
```

```
## [1] 1 1 1 1 1
```

Multinomial logistic regression

Limitations

- Larger samples required compared to simpler models, particularly when categories are imbalanced.
- Difficult to evaluate model fit.
- Unstable if some variables perfectly predict category membership or have no overlap with certain categories.

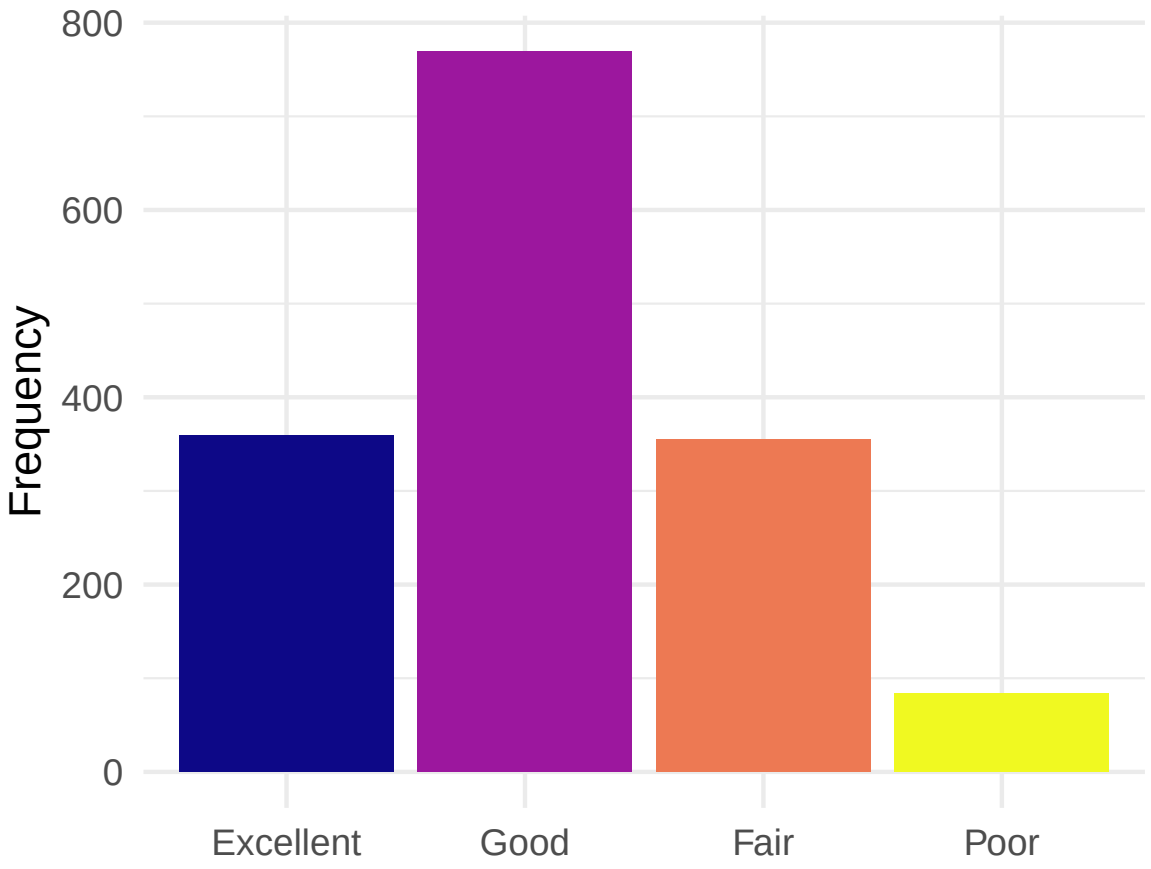
Ordinal logistic regression

- The multinomial framework could be used for ordinal data, but it ignores any information about the order of categories.
- **Ordinal** logistic regression accounts for ordering by using **cutpoints** to map the intervals between categories onto a linear scale.

Ordinal logistic regression

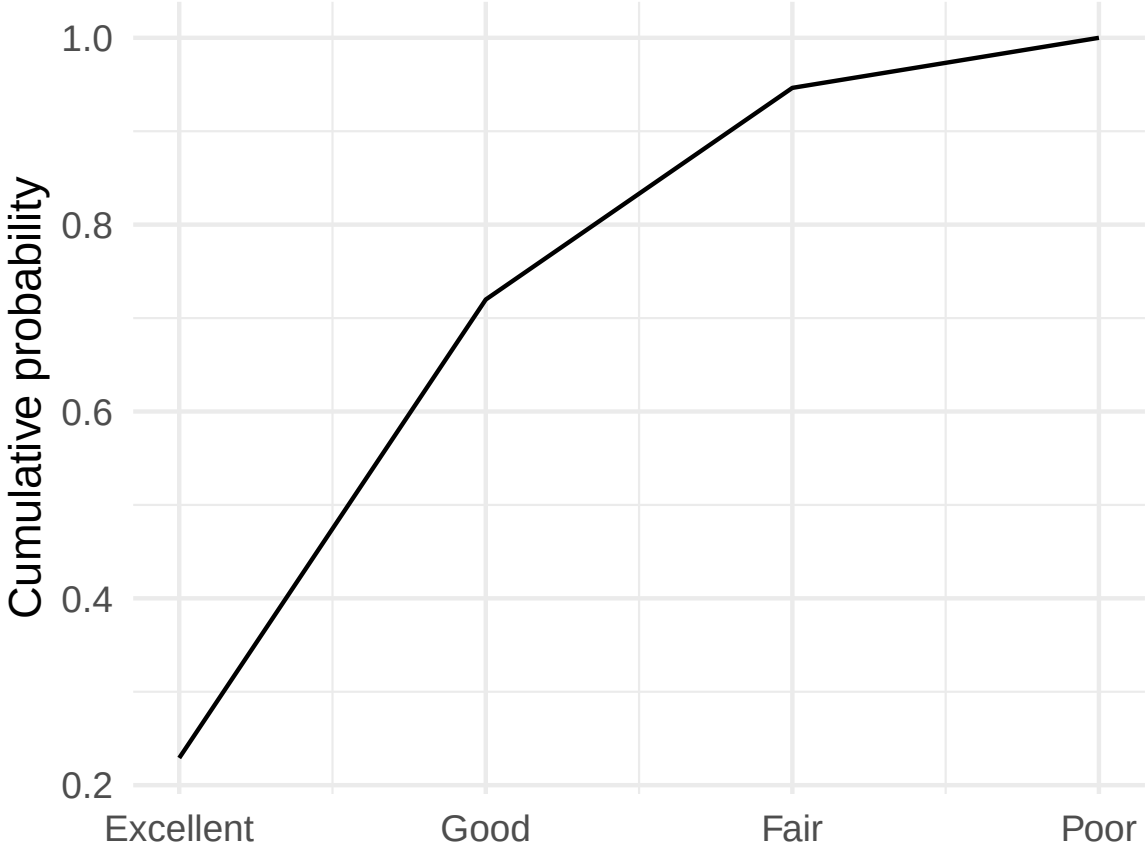
- Methodology:
  - Map categorical outcome onto cumulative probability scale using cumulative link.
  - Convert to log-cumulative-odds, analogue of the logit link for cumulative scale.
  - Construct a linear model to examine association between predictors and outcome, while maintaining information on order.

Data: Self-reported health



Ordinal logistic regression

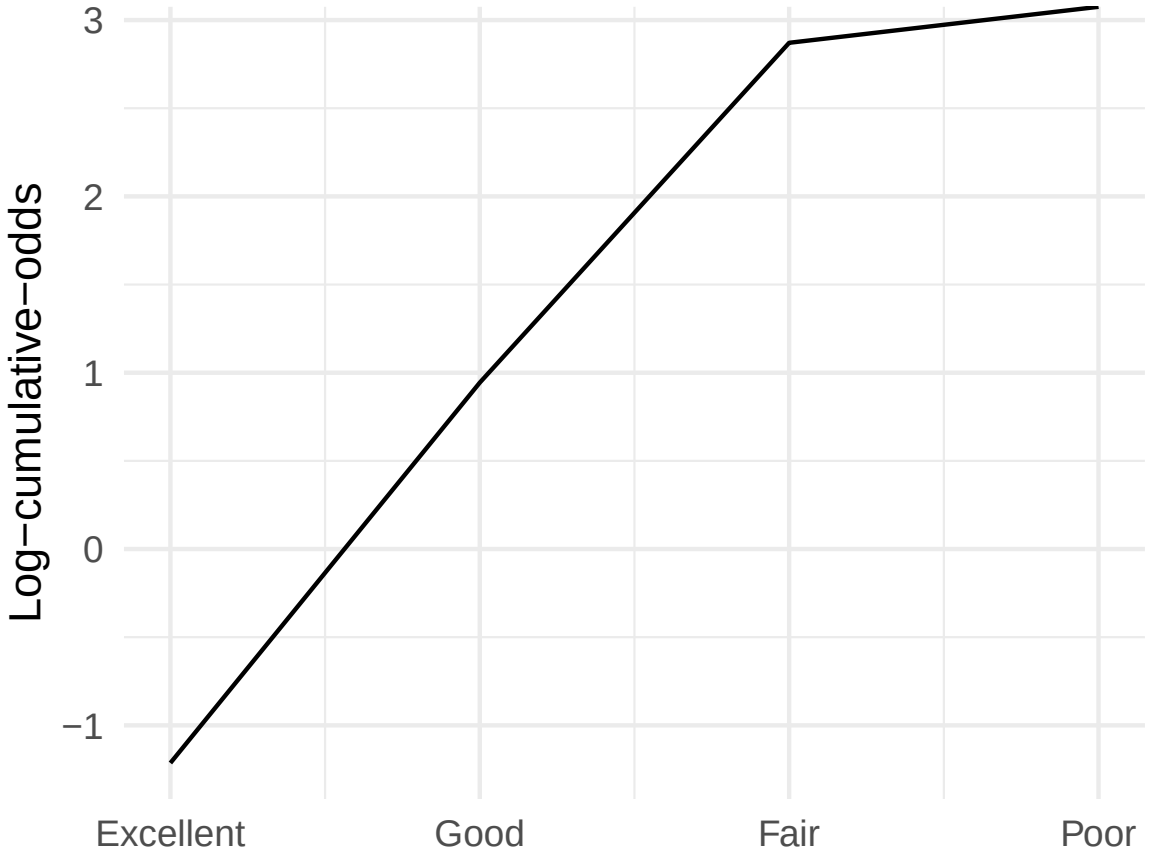
Cumulative probabilities of each class




```
## [1] 0.229 0.720 0.946 1.000
```

Ordinal logistic regression

Log cumulative odds




```
## [1] -1.213  0.944  2.871    Inf
```

Ordinal logistic regression

Estimation

- Each cutpoint represents the log-cumulative-odds that y_i is less than or equal to some value k. These are analogous to *category-specific intercepts*.

$$log(\frac{P(y_i \leq k)}{1 - P(y_i \leq k)}) = \alpha_k$$

- The intercept for the final value is ∞ since $log(\frac{1}{1-1}) = \infty$. Therefore we only need $K - 1$ intercepts.

Ordinal logistic regression

Estimation

- If we use the inverse link, we can go back from log-cumulative-odds to cumulative probabilities. The likelihood of k is expressed as

$$p_k = P(y_i = k) = P(y_i \leq k) - P(y_i \leq k - 1)$$

- In the context of this example, we could express the likelihood of "Good" health as

$$p_{\text{good}} = P(y_i = \text{good}) = P(y_i \leq \text{good}) - P(y_i \leq \text{excellent})$$

Ordinal logistic regression

Estimation

- Given this $K-1$ length vector of intercepts, $\alpha_{k \in K-1}$, we can use a linear model to predict the log-cumulative-odds that $y_i \leq k$ given a matrix of predictors X:

$$\phi_i = \beta X_i$$

$$log(\frac{P(y_i \leq k)}{1 - P(y_i \leq k)}) = \alpha_k - \phi_i$$

- Note: because ϕ_i is *subtracted*, a positive coefficient indicates that the predictor shifts the distribution toward higher outcome categories (e.g. toward poorer health). Some software and textbooks use the opposite sign convention.

Ordinal logistic regression

Estimation

- Once again, we cannot fit the model using `glm`. Instead, we can use the `polr` function from the `MASS` package.
- `rstanarm` includes a Bayesian implementation, `stan_polr`

Ordinal logistic regression

Estimation

```
library(MASS)
m2 <- polr(health ~ age + I(log(realrinc)) + educ + sex + race,
                      data = gss, Hess = TRUE)
```

The argument `Hess = TRUE` ensures the Hessian matrix is stored. This is necessary for subsequent model evaluation.

Ordinal logistic regression

	Log odds	Odds ratios
age	0.004	1.004
	(0.005)	(0.005)
I(log(realrinc))	-0.204	0.815
	(0.058)	(0.047)
educ	-0.102	0.903
	(0.024)	(0.022)
sexMale	0.098	1.102
	(0.130)	(0.144)
raceBlack	0.148	1.160
	(0.174)	(0.201)
raceOther	0.248	1.282
	(0.207)	(0.265)
Num.Obs.	906	906
AIC	1987.5	1987.5

Significance tests not provided by polr; no stars displayed.

Ordinal logistic regression

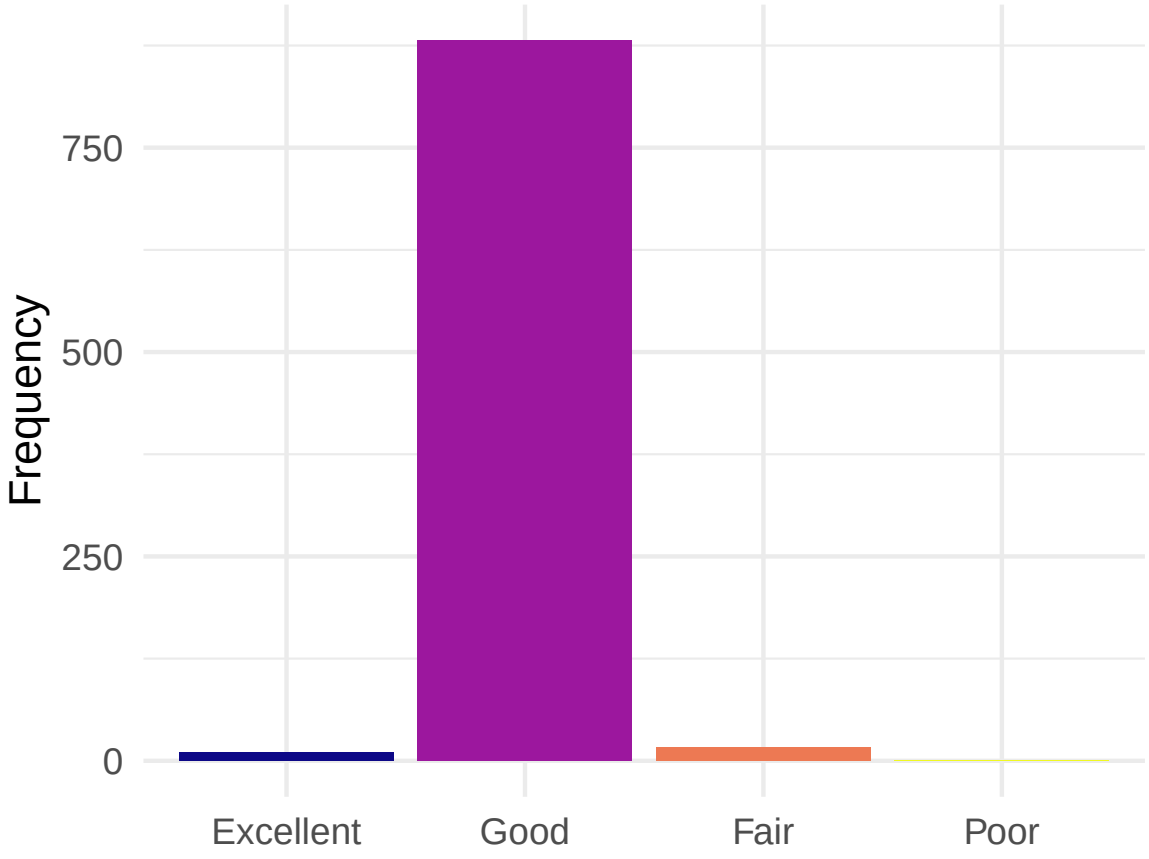
Predictions

```
preds2 <- predict(m2, gss %>% drop_na(health, age, sex,
                                      race, realrinc, educ))
preds2 %>% head(10)
```

```
##  [1] Good Good Good Good Good Good Good Good Good Good
## Levels: Excellent Good Fair Poor
```

Ordinal logistic regression

Predictions



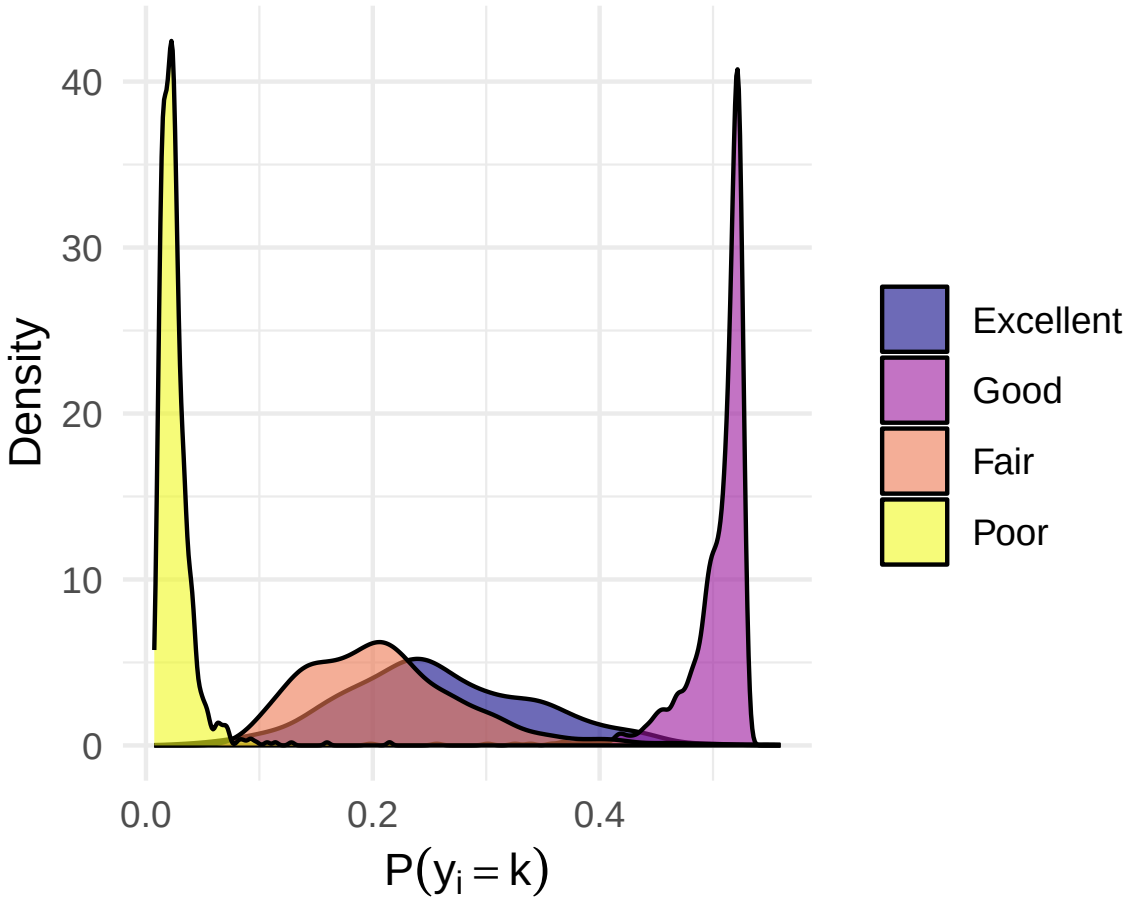
Ordinal logistic regression

Predictions

```
probs2 <- predict(m2, type = "probs",
                  gss %>%
                      drop_na(health, age, sex,
                              race, realrinc, educ))
probs2 %>% round(3) %>% head(5)
```

```
##   Excellent  Good  Fair  Poor
## 1     0.193 0.517 0.258 0.032
## 2     0.219 0.523 0.231 0.027
## 3     0.404 0.470 0.114 0.011
## 4     0.307 0.512 0.163 0.017
## 5     0.174 0.509 0.281 0.036
```

Ordinal logistic regression



Ordinal logistic regression

More predictions

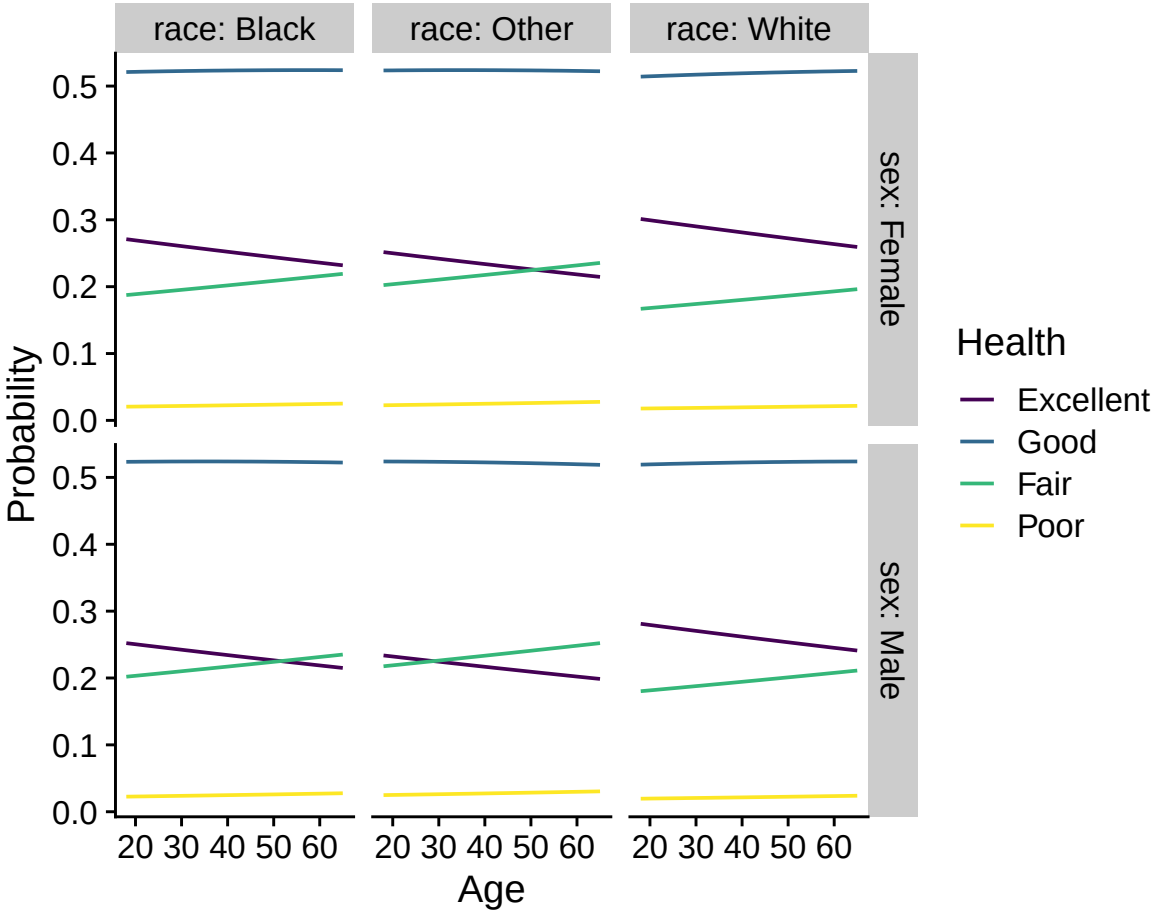
We can easily generate predictions for all combinations of predictors.

```
newdat <- expand_grid(
  race = c("Black", "White", "Other"),
  sex = c("Female", "Male"),
  educ = 12,
  realrinc = c(50000),
  age = 18:65)

newpreds <- predict(m2, newdat, type = "probs")
head(newpreds, 5) %>% round(3)
```

```
##   Excellent  Good  Fair  Poor
## 1     0.271 0.521 0.187 0.021
## 2     0.270 0.521 0.188 0.021
## 3     0.269 0.521 0.189 0.021
## 4     0.268 0.521 0.189 0.021
## 5     0.268 0.522 0.190 0.021
```

Ordinal logistic regression



Ordinal logistic regression

Cutpoints

The cutpoints can be extracted using the `zeta` parameter.

```
cuts <- m2$zeta
print(cuts)
```

```
## Excellent|Good      Good|Fair      Fair|Poor
##     -4.1986283     -1.8720014      0.6567534
```

Ordinal logistic regression

Cutpoints

We can obtain the cumulative probability at each cutpoint by using the inverse logit function, $\frac{e^x}{1+e^x}$.

```
inv.logit <- function(x) {
    return(exp(x) / (1 + exp(x)))
    }

cut.probs  <- inv.logit(cuts)
cut.probs %>% round(3) %>% print()
```

```
## Excellent|Good      Good|Fair      Fair|Poor
##          0.015          0.133          0.659
```

Ordinal logistic regression

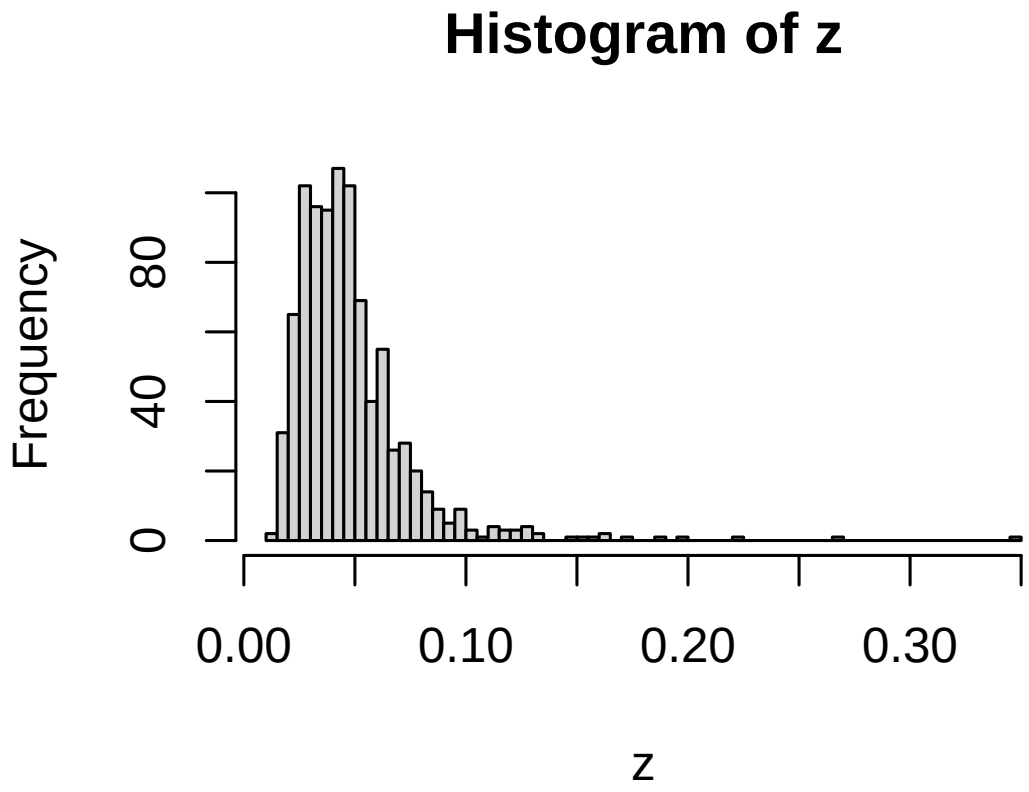
Latent variables

The ordinal model implies a *latent variable* underlying the observed categories. We can approximate each observation's position on the cumulative probability scale by applying the inverse logit to the linear predictor (ϕ_i), without subtracting the cutpoints. We can then observe where each observation falls between the cutpoints.

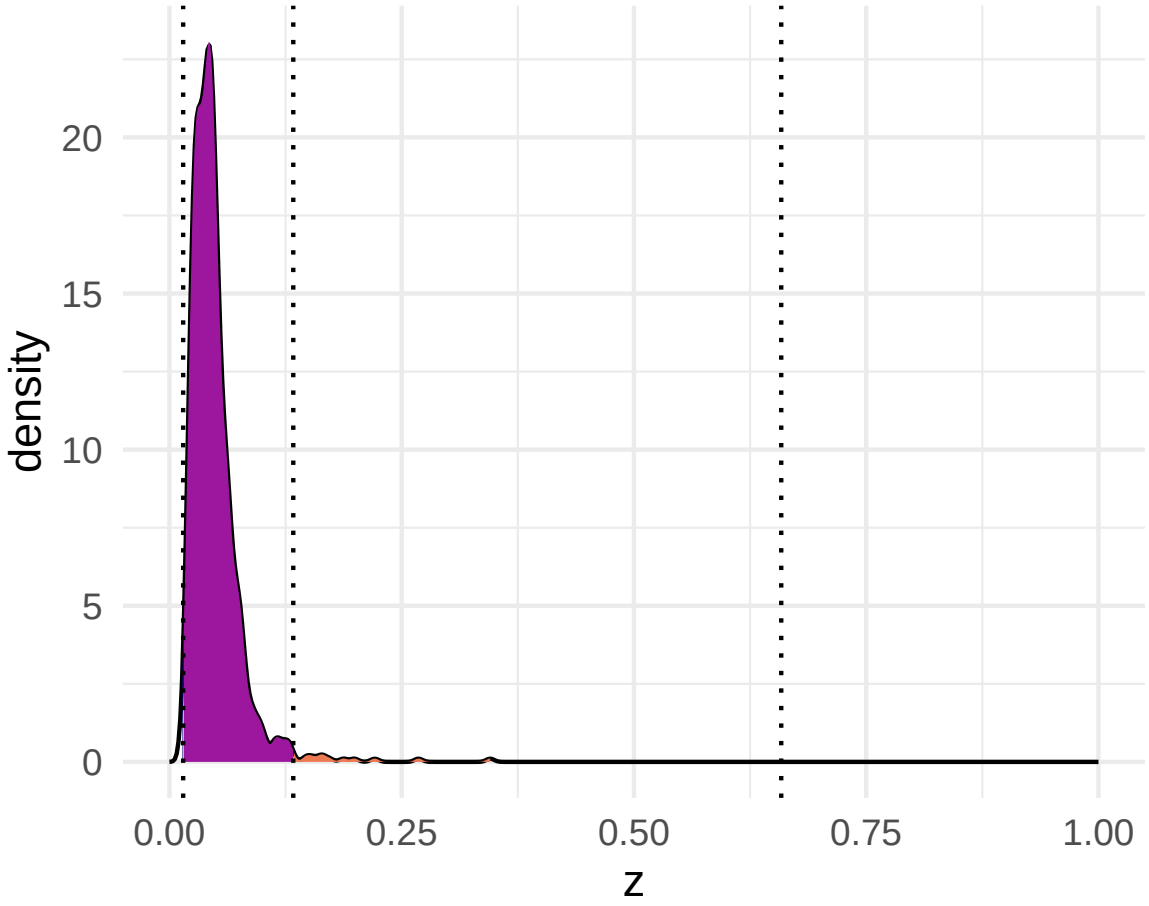
```
z <- m2$lp  %>% inv.logit()
z %>% head(10) %>% round(3)
```

```
##     7     8    11    14    16    19    21    24    25    27
## 0.059 0.051 0.022 0.033 0.067 0.018 0.033 0.048 0.038 0.051
```

Ordinal logistic regression



Ordinal logistic regression



Ordinal logistic regression

Limitations

- Similar to multinomial logistic regression
  - Larger samples required compared to simpler models.
  - Difficult to evaluate model fit.
  - Unstable if some variables perfectly predict category membership or have no overlap with certain categories.

Ordinal logistic regression

Proportional odds assumption

- Assumes the effect of each predictor on the log-odds of being at or below any given category is the same regardless of where the outcome is split (hence one set of coefficients).
- Additional tests are required to verify this is met.[3]

[3] See the UCLA stats blog for details.

Categorical outcomes

Frequentist and Bayesian approaches

- Due to the complexity of the models, many frequentist approaches require additional testing and analysis to diagnose issues and assess model fit.
- In contrast, we can use the same tools to evaluate Bayesian models:
  - Trace plots and MCMC diagnostics for estimation issues
  - LOO-CV and ELPD for fit
  - PSIS diagnostics for outliers
  - Posterior predictive checks for predictions and fit
- Either way, these models are more cumbersome to work with than other single-equation GLMs.

Summary

- Categorical outcomes can be modeled using specialized types of generalized linear models.
- Unordered categories
  - Multinomial logistic regression
- Ordered categories
  - Ordinal logistic regression
  - OLS if many categories and equal intervals
- These models are complex and more difficult to fit and interpret than previous models we have covered.

Next week

- Grouped and nested data
  - Clustered standard errors
  - Fixed effects
  - Random effects
- Autocorrelation
- Project workshop